CHARACTERIZATIONS OF FINITE CENTRAL SIMPLE ALGEBRAS

THE STACKS PROJECT AUTHORS

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Editorial note. The proof is the author's complete argument; each of the six conditions is accounted for in the circle of implications, the only steps attributed elsewhere being named results of the same chapter carried on this card as navigation stubs.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Finite central simple algebras. Same chapter, Section section-algebras and the definitions of a finite central simple algebra. Supplies the equivalence of the first two conditions as a consequence of the definitions.
- Existence of a separable splitting field. Same chapter, Proposition proposition-separable-splitting-field. Supplies that a finite central simple algebra has a separable splitting field, whose Galois closure is then also a splitting field, giving the chain to the matrix algebra conditions.
- Wedderburn's theorem. Same chapter, Theorem theorem-wedderburn. Supplies the description of a finite simple algebra as a matrix algebra over a finite skew field, which is the last condition.
- The matrix algebra case. Same chapter, Lemma lemma-matrix-algebras. Supplies that the base change of a matrix algebra is again central simple, which is used to return from the third condition to the first.

Source reference. Original section 073Y: sources/stacks-brauer/pinned-original-source.tex, source lines 247–247. Navigation pointer invoked at proof line 1006; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:brauer.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 0752: sources/stacks-brauer/pinned-original-source.tex, source lines 923–923. Navigation pointer invoked at proof line 1008; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074D: sources/stacks-brauer/pinned-original-source.tex, source lines 477–477. Navigation pointer invoked at proof line 1014; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 0747: sources/stacks-brauer/pinned-original-source.tex, source lines 358–358. Navigation pointer invoked at proof line 1017; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let k be a field. For a k-algebra A the following are equivalent*

1. *A is finite central simple k-algebra,*
2. *A is a finite dimensional k-vector space, k is the center of A, and A has no nontrivial two-sided ideal,*
3. *there exists $d \geq 1$ such that $A \otimes_k \bar{k} \cong Mat(d \times d, \bar{k})$,*
4. *there exists $d \geq 1$ such that $A \otimes_k k^{sep} \cong Mat(d \times d, k^{sep})$,*
5. *there exist $d \geq 1$ and a finite Galois extension k'/k such that $A \otimes_k k' \cong Mat(d \times d, k')$,*
6. *there exist $n \geq 1$ and a finite central skew field K over k such that $A \cong Mat(n \times n, K)$.*

The integer d is called the degree of A.

Proof. The equivalence of (1) and (2) is a consequence of the definitions, see Section 073Y. Assume (1). By Proposition 0752 there exists a separable splitting field $k \subset k'$ for A. Of course, then a Galois closure of k'/k is a splitting field also. Thus we see that (1) implies (5). It is clear that (5) $\Rightarrow$ (4) $\Rightarrow$ (3). Assume (3). Then $A \otimes_k \overline{k}$ is a finite central simple $\overline{k}$-algebra for example by Lemma 074D. This trivially implies that A is a finite central simple k-algebra. Finally, the equivalence of (1) and (6) is Wedderburn's theorem, see Theorem 0747. □